

BeerChat

Real-time, venue-level social presence with double opt-in meetups

CS 415/515 Software Engineering — Project Milestone 2 (Test It)

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Repository	github.com/Arcce9/BeerChat
Git tag / release	`milestone-2` · release `v0.2.0` (previous: `milestone-1` / `v0.1.0`)
Pull requests	#1 features + tests · #2 security · #3 E2E, CI, docs · #4 review follow-up
Submitted	November 2026

1 · Expanded working application & repository

Repository: https://github.com/Arcce9/BeerChat, tag `milestone-2`. BeerChat is a mobile app (React Native / Expo) on a hosted Supabase backend; it is run locally in the iOS Simulator (§5) — there is no separate public URL until deployment in Milestone 3. The hosted backend the app talks to is live.

The Milestone 1 core loop still works end to end — `sign up → profile → pin a venue → check in → discover → mutual "open to meet" → chat` — and is now covered by automated tests at four levels, hardened against three real vulnerabilities, and expanded with two backlog features.

2 · New features completed since Milestone 1

Item	What the user gets	How it is built
BC-08 Interest-ranked people list	People at your venue are ordered by how many interests you share ("2 in common"); shared interests are highlighted.	Pure ranking function (`rankBySharedInterests` : shared count → already-here before heading-over → name), unit-tested.
BC-10 Ephemeral chat	Every match and its chat disappear 6 hours after matching; the time is shown ("disappears at 12:47 AM").	`matches.expires_at` ; RLS hides expired matches/messages instantly; a `pg_cron` job purges them every 15 min; a client timer closes an open chat at expiry.
SEC-1…3 Security hardening	Only people at the same venue can see who is there (others see anonymous counts); requests only between visible people at the venue; server-side input limits.	Migration `…000009` : RLS policies, `venue_counts()` aggregate RPC, check constraints (§4).
Quality fixes	"List"/"Sign out" no longer hidden under the iOS status bar; no profile carry-over when switching accounts; profile setup enforces the specified 3–5 interests.	Found by the E2E run and by ESLint respectively (§3.3).
Demo & tooling	One-device demo with a populated venue; CI on every PR.	`npm run seed:demo` ; GitHub Actions.

Scope note (agile change, small magnitude): the Socket.io presence server (BC-09) and the OpenAI-embedding upgrade of ranking move to Milestone 3 — both are deployed services, which is that milestone's theme; BC-08 ships now with exact shared-interest ranking.

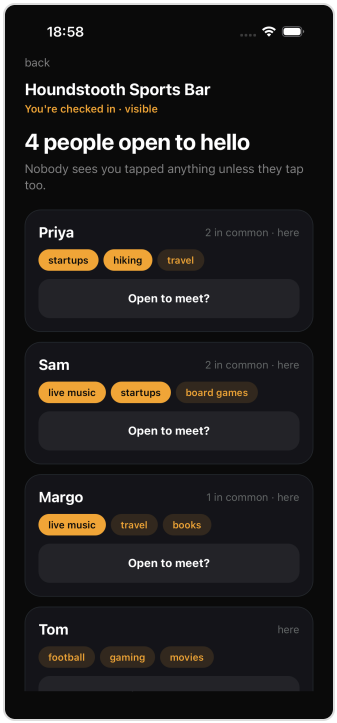
2.1 Consistency with Milestone 1 and the proposal

What the proposal's milestone plan ("Milestone 2: Socket.io integration, AI interest ranking, security controls, test suite") and the Milestone 1 report ("Next (M2)", known limitations) committed to, and where each item stands:

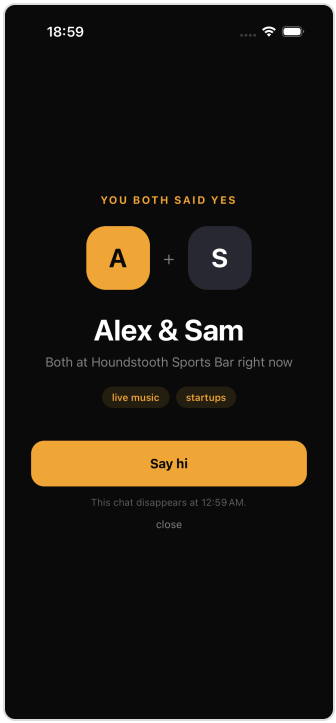
Commitment	Source	Status at Milestone 2
Test suite	Proposal M2 · M1 "Next"	**Done — unit, component, integration, E2E (§3)**
Security controls	Proposal M2	**Done — SEC-1…3 (§4)**
Time-based chat expiry (BC-10)	M1 "Next" · M1 known limitations	**Done — 6 h expiry**
CI	M1 "Next" (proposal: M3)	**Done early — GitHub Actions on every PR**
Interest-ranked list (BC-08) using OpenAI embeddings + pgvector	Proposal M2 · M1 §5.1, "Next"	**Partly** — ranking ships using exact shared-interest overlap; embedding similarity (so "hiking" and "climbing" count as related) is not yet built
Socket.io presence service (BC-09, T-05)	Proposal M2 · M1 §5.1 and ADR-0001	**Not done** — presence still runs on Supabase Realtime (works, verified live on two devices); moved to M3 with the other deployed services
Geofenced check-in	M1 known limitations	**Not done** — check-in is still a manual "I'm here" button; moved to M3
Milestone 1 workflow keeps working	M2 instructions	**Verified — the integration suite and E2E flows exercise the full M1 loop on the M2 build**

2.2 Design changes since Milestone 1

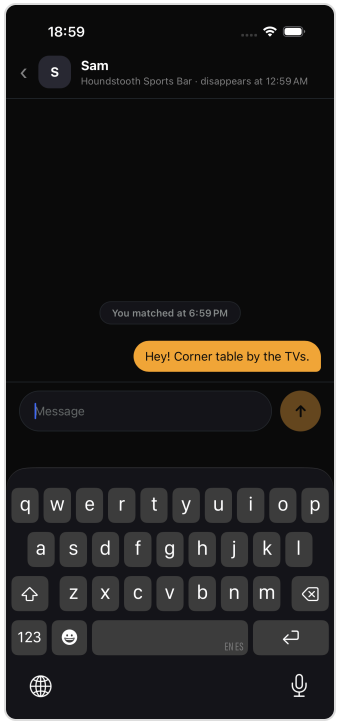
- **API contract** (M1 §5.3 table): others' `pins` / `profiles` are now readable only by viewers checked in and visible at the same venue; venue counts come from the new `venue_counts()` RPC; `meet_requests` inserts require the requester to be checked in and visible at the venue and the target to be visible there; `matches` / `messages` readable only until `expires_at`. Recorded as **ADR-0003** (`docs/adr/0003-venue-scoped-visibility.md`).
- **Data model** (M1 Fig. 2): `MATCH` gains `expires_at`; `PROFILE` gains length/count constraints. Entities and relationships are otherwise unchanged.
- **Modules** (M1 §5.3): new `lib/logic.ts` (pure, unit-tested logic used by the screens); the boundary rule "only MapHome owns realtime subscriptions and mutation state" still holds.
- **Process** (M1 §3): the same incremental, feature-gated model, now with pull requests, CI and the QA checklist added to the Definition of Done (`docs/QA-PLAN.md` §4).



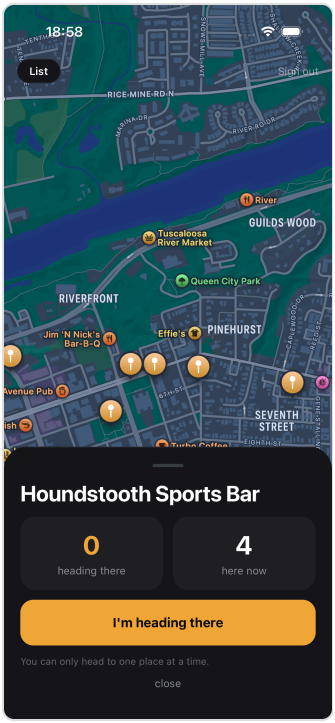
BC-08 — ranked by shared interests, shared tags filled amber



Match screen with BC-10 expiry note



Chat header: "disappears at …" (BC-10)



Venue sheet: anonymous counts (SEC-1)

3 · Testing and quality assurance

3.1 Test levels (full plan: `docs/QA-PLAN.md`)

Level	Tool	Scope	Result
Static	TypeScript strict, ESLint	whole app	**0 type errors, 0 lint errors**
Unit	Jest	`logic.ts`: ranking, expiry, validation, pin state machine (32 tests)	**pass**
Component	Jest + React Native Testing Library (Supabase mocked)	MapHome, PeopleHere, MatchScreen, ChatScreen (22 tests)	**pass**
Integration	Jest in Node against the **live** Supabase project, 3 real accounts	auth, RLS privacy rules, match/unmatch triggers, realtime delivery, expiry, security (30 tests)	**30 / 30 pass**
End-to-end UI	Maestro on iOS Simulator	full journey through the real UI (2 flows)	**pass on 3 screen sizes**

Integration tests for the main user flows (`app/__tests__/integration/flows.test.ts`) exercise the real system, not mocks: wrong password rejected; signed-out clients see nothing; a visible person appears to someone at the same venue and disappears when hidden; a one-sided request is invisible to its target; the return request creates exactly one match visible to both and to no outsider; clients cannot create matches directly; the match is set to expire 6 h after creation; **a message is delivered to the other participant over Realtime**; a message cannot be sent under someone else's name; outsiders cannot read the chat; withdrawal dissolves the match and deletes the chat.

3.2 Coverage report and target

File	Statements	Branches	Functions	Lines
`lib/logic.ts`	100 %	94.3 %	100 %	100 %
`MatchScreen.tsx`	100 %	100 %	100 %	100 %
`ChatScreen.tsx`	93.1 %	83.3 %	100 %	92.3 %
`PeopleHere.tsx`	82.1 %	73.5 %	92.9 %	83.3 %
`MapHome.tsx`	76.6 %	71.0 %	70.8 %	79.1 %
All files	**83.7 %**	**77.9 %**	**83.0 %**	**84.9 %**

> **Target (enforced — the build fails below it):** pure logic ≥ 95 % statements / 90 % branches / 100 % functions; logic and components (`src/lib`, `src/components`; the sign-in and profile screen in `src/app` is not measured) ≥ 80 % statements & lines, 70 % branches, 75 % functions. **Justification:** pure logic is cheap to test exhaustively and is where ranking/expiry/validation bugs hide, so it gets near-total coverage. The uncovered remainder of the screens is almost entirely Supabase realtime callbacks and network-error branches; mocking those would only test the mock. Those paths are instead covered against the real

> backend by the 30 integration tests and the E2E flows, which line-coverage tools cannot count. HTML report committed at `docs/evidence/coverage/index.html` (regenerate with `npm run test:coverage`); CI also uploads it as an artifact on every run.

3.3 Quality-assurance evidence

- **QA checklist** used on every PR (also the PR template): CI green · tests for new logic/UI · privacy enforced in the DB with an integration test · new migration files only · checked on small and large iPhones · bugs logged.
- **Defects found by the QA process this milestone** (all fixed, logged in `docs/BUGLOG.md` / commit history):
  - ESLint `react-hooks/set-state-in-effect` → a previous account's profile could carry over after switching accounts → profile state keyed by user id.
  - E2E run → "List" and "Sign out" rendered under the iOS status bar and swallowed taps → layout fix.
  - E2E on the small iPhone 16e → keyboard covers the password field (documented; `returnKeyType="next"` planned).
  - Integration suite → three security vulnerabilities (§4).
  - Code review → SEC-1 could be bypassed with a hidden pin → viewers must be checked in and visible (migration `…000010`, §4).
  - CI on the merged code → the first ChatScreen unit test hit Jest's 5 s timeout in 2 of 3 runs → default test timeout raised to 15 s.
  - Requirements review → profile setup accepted 1 interest although BC-02 specifies 3–5 → fixed.
- **CI:** GitHub Actions runs typecheck, lint and unit tests with coverage on every pull request (first run: green, 1 m 29 s); integration tests run on `main` and on demand.

4 · Code review, cross-platform testing and security

4.1 Peer code review

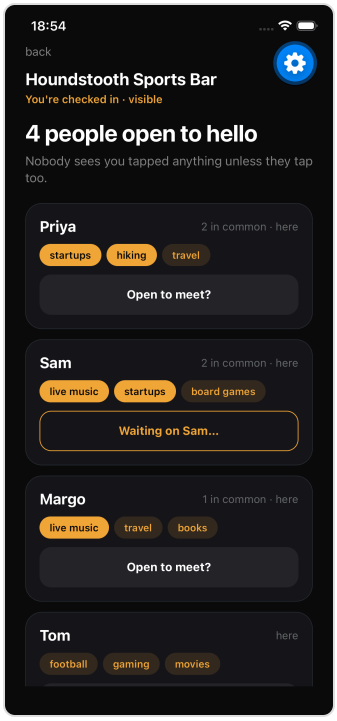
All Milestone 2 work went through pull requests, each with a description, test notes and the QA checklist. PRs #1–3 were merged before anyone had reviewed them; Karthik Gaur reviewed them afterwards on GitHub. PR #4 (Arda Alici's SEC-1 follow-up plus the final fixes) was reviewed before it merged, and `milestone-2` is tagged on the merged result, so the PR pages are the review record:

PR	Scope	Review
#1	BC-08, BC-10, logic extraction, unit/component tests, lint fix	Karthik Gaur (@Karthikgaur8), after merge
#2	SEC-1…3 + integration suite (test-first: failing tests committed before the fix)	Karthik Gaur, after merge; Arda Alici (@Arcce9) found the SEC-1 bypass fixed in #4
#3	E2E flows, status-bar fix, CI, QA/security/contributing docs	Karthik Gaur, after merge
#4	SEC-1 bypass fix and 7 more integration tests (Arda Alici), CI test-timeout fix, final report	Karthik Gaur, before merge (Arda's commits)

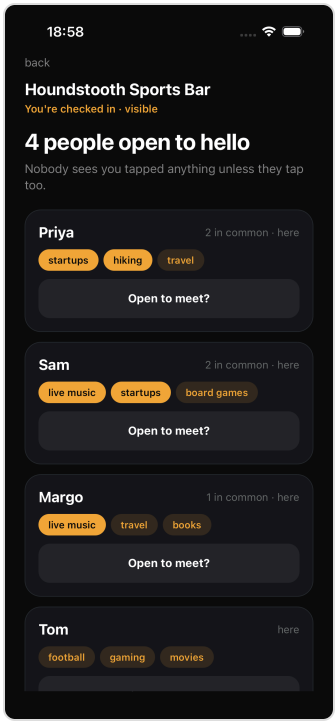
Branching and commit conventions (`docs/CONTRIBUTING.md`): short-lived `feature/`, `fix/`, `security/`, `docs/` branches off an always-runnable `main`; merge only via reviewed PR with green CI; Conventional Commits (`feat:`, `fix(security):`, `test:`, `ci:`…) referencing backlog/security ids; AI-assisted commits carry a `Co-Authored-By` trailer naming the model. **Changelog and releases:** `CHANGELOG.md`; tagged releases `v0.1.0` (Milestone 1) and `v0.2.0` (Milestone 2).

4.2 Cross-viewport testing

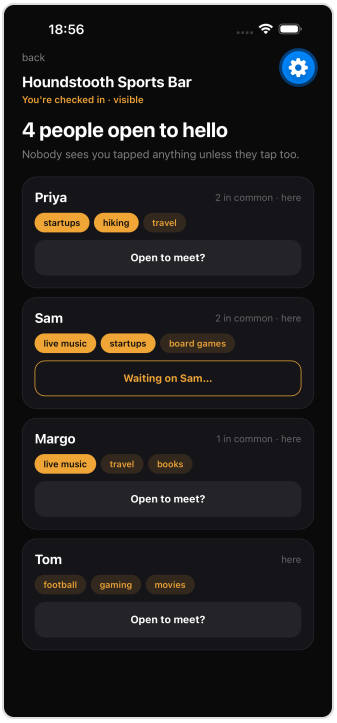
The Maestro flow `01-signin-checkin-discover` was run unchanged on the smallest and largest current iPhones and on the iPhone 17; the full journey (flows 01 + 02) on the iPhone 17. Android was not run this milestone (team decision); the app uses only cross-platform React Native components.



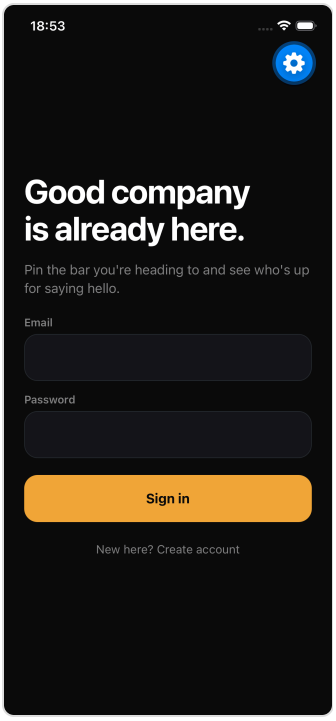
iPhone 16e (390 × 844 pt) ✓



iPhone 17 (402 × 874 pt) ✓



iPhone 17 Pro Max (440 × 956 pt) ✓



iPhone 16e sign-in ✓

4.3 Threat model (main attack surfaces)

The phone talks to the database directly with a publishable key that every copy of the app contains, so **the database is the trust boundary**: any rule that exists only in the UI is not a control. Full STRIDE table: `docs/SECURITY.md`.

Surface	Threat	Before M2	Control
pins / profiles reads	enumerate who is visible at every venue (real-time whereabouts)	**vulnerable**	SEC-1: same-venue RLS; counts via aggregate RPC
meet_requests insert	request arbitrary users from anywhere, incl. hidden people	**vulnerable**	SEC-2: RLS WITH CHECK on presence + visibility
profiles update	store huge names/interest lists shown to others	**vulnerable**	SEC-3: DB check constraints
meet_requests reads	learn who said yes to you	**mitigated (M1)**	requester-only RLS
matches / messages	forge matches, spoof sender, read others' chats, linger	**mitigated**	trigger-only matches; sender = self; participant RLS; 6 h expiry

4.4 OWASP Top 10 (2021) review

Category	Finding
A01 Broken Access Control	**Two real issues (SEC-1, SEC-2) — fixed.** RLS on every table; access rules asserted by integration tests. Hiding chats at expiry and the purge job need time to pass, so no automated test covers them yet.
A02 Cryptographic Failures	Supabase Auth: bcrypt, TLS, JWT. No custom crypto.
A03 Injection	PostgREST parameterises queries; no string-built SQL; DB functions pin `search_path = ''`.
A04 Insecure Design	**Limits existed only in the UI (SEC-3) — moved into DB constraints.**
A05 Security Misconfiguration	RLS everywhere; purge function's execute revoked from clients; email confirmation off for the demo (documented).
A06 Vulnerable Components	`npm audit` reviewed; Expo SDK patches applied; CI installs from the lockfile.
A07 Identification & Auth	Supabase sessions; profile-carry-over bug on account switch fixed.
A08 Integrity · A09 Logging · A10 SSRF	Lockfile + versioned migrations · Supabase request logs (alerting in M3) · N/A (no server-side fetches).

4.5 Implemented security control — before / after evidence

Test-first: the security tests (asserting the *secure* behaviour) were committed and run **before** the fix (`9d34f1c`, log `docs/evidence/security-BEFORE.txt`), then the migration was applied and the same suite re-run (`a46267d`, `security-AFTER.txt`).

Security test	Before	After
SEC-1 outsider cannot list who is checked in at a venue	✗ **received 2 user ids**	✓
SEC-1 outsider cannot read profiles of people at a venue	✗	✓
SEC-1 someone at a different venue cannot list this one	✗	✓
SEC-1 anyone signed in still gets anonymous counts	✗ **RPC did not exist**	✓
SEC-2 outsider cannot send a meet request	✗ **accepted**	✓
SEC-2 request cannot target a hidden person	✗ **accepted**	✓
SEC-3 rejects a 5,000-character display name	✗ **accepted**	✓
SEC-3 rejects more than 5 interests	✗ **accepted**	✓
Whole integration suite	**8 failed · 15 passed**	**23 passed**

How the TA observes the control: (1) `npm run test:integration` — SEC-1…3 pass; (2) in the app, open a venue *before* checking in — counts are visible, identities are not; after checking in visibly at that venue, the people list appears.

5 · Updated setup, run and verification instructions

Unchanged prerequisites: macOS with Xcode 16+ (iOS Simulator runtime), Node.js 20+. Full instructions: `README.md`; click-by-click script with expected results: `docs/VERIFICATION.md`.

```
git clone git@github.com:Arcce9/BeerChat.git && cd BeerChat && git checkout milestone-2
cd app && npm install          # dependencies (Expo SDK 57, Jest, Testing Library)
npm run seed:demo              # sample data: Sam, Priya, Tom checked in at Houndstooth Sports Bar
npx expo start                 # press i → iOS Simulator (Expo Go installs automatically)

npm run typecheck && npx expo lint
npm test                       # 54 unit + component tests
npm run test:coverage          # + coverage report and enforced thresholds
npm run test:integration       # 30 tests against the live backend
npm run e2e                    # optional: Maestro UI flows (brew install maestro)
```

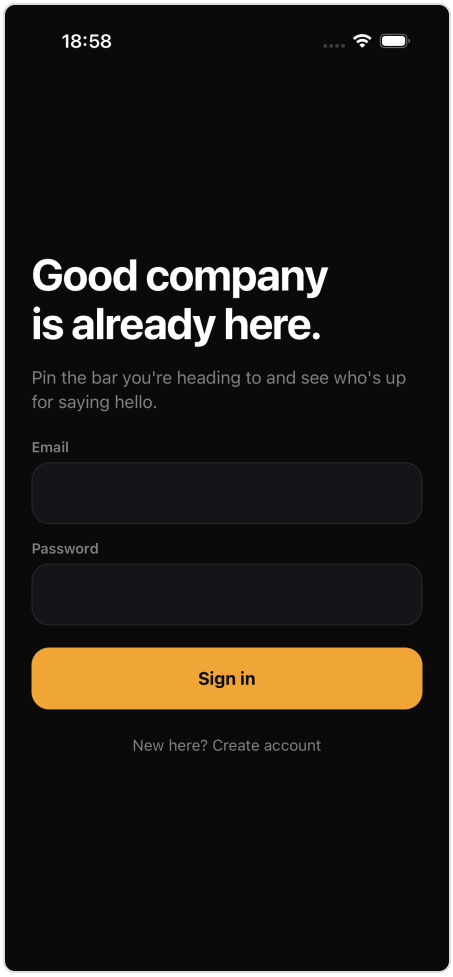
Environment: `app/.env` (Supabase URL + publishable key, safe to ship by design — all data protected by RLS) and `app/.env.test` (integration-test account password) are committed, so no configuration is needed; `app/.env.example` documents the variables. **Migrations:** ten ordered files in `supabase/migrations/`, already applied to the hosted database (`supabase db push` for a fresh project); M2 adds `…000008_chat_expiry`, `…000009_security_hardening` and `…000010_viewer_must_be_visible`. **Demo accounts:** `ta.demo1@beerchat.dev` / `BeerChat-M1-demo1` (Alex — sign in as this one); demo2–4 (Sam, Priya, Tom) follow the same pattern.

#	TA checks	Expected
1	Milestone 1 workflow (two simulators, two demo accounts): check in → see each other live → double opt-in → chat → withdraw	as in M1; still works
2	As Alex, open Houndstooth *before* checking in	anonymous counts, no identities (security control)
3	Check in, go visible, "See who's open to hello"	Priya and Sam ("2 in common") above Tom (BC-08)
4	`node scripts/seed-demo.mjs sam-says-yes`, tap "Open to meet?" on Sam, "Say hi"	match + chat show "disappears at …", 6 h ahead (BC-10)
5	Run the automated test commands above	all pass

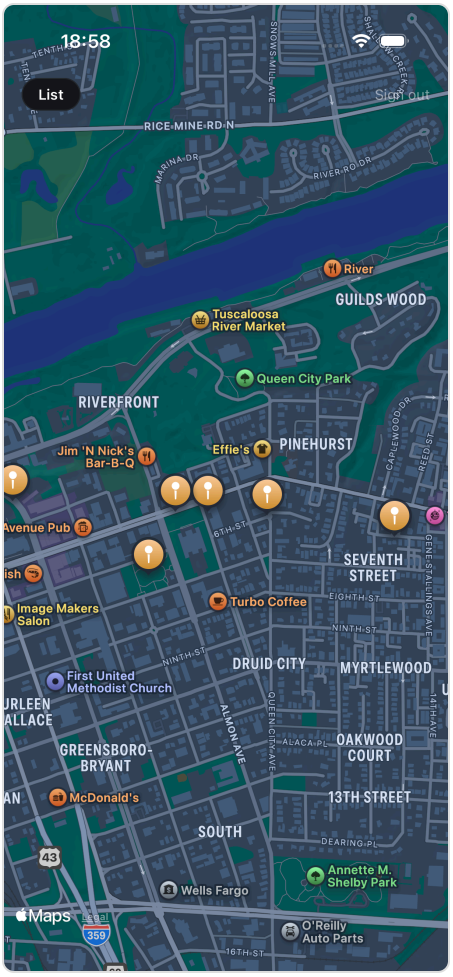
Free-tier note: the hosted Supabase project pauses after ~1 week idle; if the app shows network errors, allow 2–3 minutes for it to wake, or contact the team.

6 · Working demonstration (existing + new features)

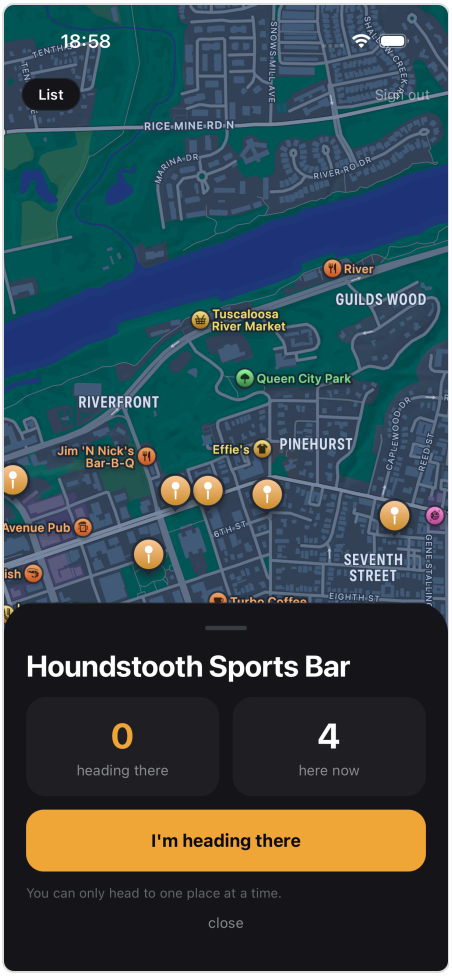
Recorded by the automated E2E flows (`app/e2e/`) on the iPhone 17 — the same steps a live demo follows:



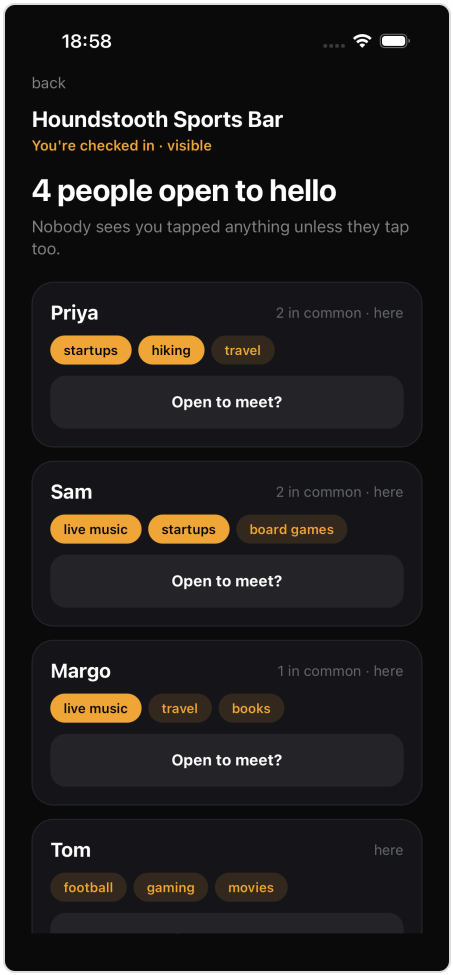
1 Sign in (M1)



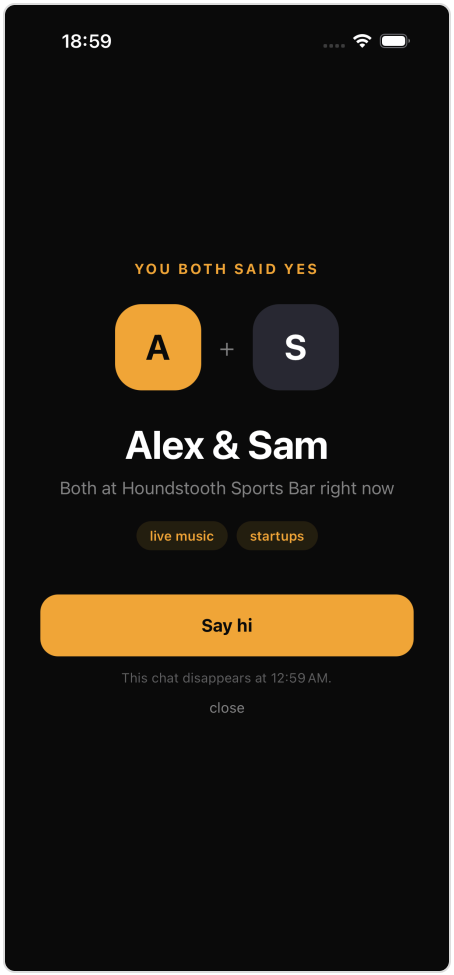
2 Map, List fallback (M1; top bar fixed in M2)



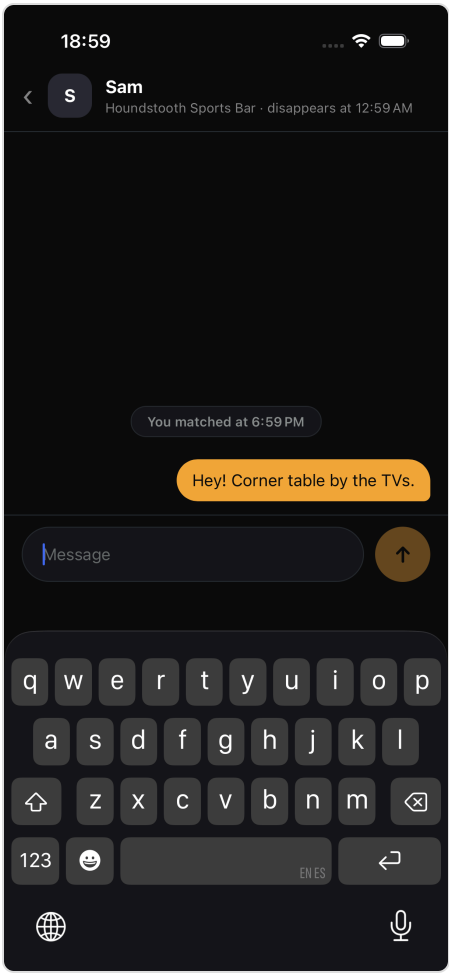
3 Venue sheet, anonymous counts (M2)



4 Ranked people (M2 BC-08)



5 Double opt-in match (M1) + expiry (M2)



6 Realtime chat (M1) with expiry (M2)

Known limitations at Milestone 2

iOS-verified only (Android not run this milestone) · check-in is manual, not geofenced, so a script can still move a visible pin between venues to list who is there (SEC-1 blocks silent viewing, not this) · presence uses Supabase Realtime rather than the planned Socket.io service · BC-08 ranks by exact shared interests, not embeddings · email confirmation disabled for the demo · free-tier database may need 2–3 minutes to wake.

7 · Contribution statements

Member	Milestone 2 contribution
Margulan Baizhakyp	Built the Milestone 2 work using a disclosed AI-assisted workflow (Claude Code; commits carry a `Co-Authored-By` trailer): BC-08 ranking, BC-10 chat expiry, the unit/component/integration/E2E test suites and CI, the SEC-1…3 security fixes with before/after evidence, the QA plan, security, contributing and changelog documents, and this report.
Arda Alici	Worked with Margulan Baizhakyp on the E2E flows, CI, documentation and report, and merged PRs #1–3 into `main`. With AI assistance (Claude Code), reviewed the Milestone 2 code, found that SEC-1 could be bypassed with a hidden pin, and wrote the fix (migration `…000010`) with regression tests, added a CI timeout and forced exit for the integration tests, added five more access-rule tests, and completed the Milestone 2 bug log.
Karthik Gaur	With AI assistance (Claude Code), reviewed PRs #1–3 on GitHub, found and fixed a unit test that was failing CI on the merged code, reviewed and merged Arda's SEC-1 follow-up (PR #4), tagged `milestone-2` and published release `v0.2.0`, and finalized this report and the README.